Synthetic vs. Translated Training Data for Multilingual Aspect-Based Sentiment Analysis

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Abstract

We build and evaluate a multilingual Aspect-Based Sentiment Analysis (ABSA) system that extracts structured opinion triplets from hotel and restaurant reviews in English and Spanish. Four mT5-small models are trained under different data conditions and compared against a zero-shot Mistral-7B baseline. The central question is whether LLM-generated synthetic Spanish training data can substitute for machine-translated data. On the hotel split, synthetic data alone performs poorly ($F_1 = 16.33$), but combined with translated data it produces the best Spanish result ($F_1 = 46.84$). The restaurant split shows the same pattern. Zero-shot Mistral-7B scores below all fine-tuned models, confirming that fine-tuning a small model on in-domain data outperforms zero-shot prompting of a much larger one.

1 Introduction

Platforms such as Booking.com and TripAdvisor ask reviewers to rate specific categories using structured score fields. This captures a fixed set of dimensions but ignores the richer signal in the free-text body of the review, where guests describe their experience in their own words.

ABSA applies machine learning to that unstructured text, identifying which aspect a reviewer is commenting on and whether their sentiment is positive, negative, or neutral. The output is a set of structured opinion triplets extracted directly from natural language, for example:

"The bathroom was disgusting but the staff were helpful."

→ `(bathroom, rooms_cleanliness, neg)`
→ `(staff, service_general, pos)`

Each triplet contains an *aspect term* (the word in the review), an *aspect category* (a standardised label), and a *sentiment polarity*.

The research question is whether LLM-generated synthetic Spanish training data can replace machine-translated Spanish. This matters for building multilingual ABSA systems in languages where labelled data is scarce. We test the same pipeline on a restaurant domain to check whether the findings hold beyond hotels.

2 Related Work

Aspect-Based Sentiment Analysis. ABSA was formalised through the SemEval shared tasks Pontiki et al. [2014, 2016], which introduced the (aspect term, aspect category, sentiment polarity) annotation scheme used here. Yan et al. [2021] introduced a joint generative T5-based framework that became the standard approach. Our system follows this seq2seq paradigm using mT5-small.

Multilingual pre-training and cross-lingual transfer. Conneau et al. [2020] showed that training on large multilingual corpora yields representations that transfer across languages without parallel data. mT5 Xue et al. [2021] applies the T5 text-to-text framework to 101 languages. We fine-tune mT5-small for ABSA and evaluate how well that transfer holds for Spanish.

Multilingual ABSA. M-ABSA Wu et al. [2025] extended ABSA to 21 languages by machine-translating the original English SemEval annotations. This is the dataset we build on. Rather than treating machine translation as the only route to multilingual training data, we substitute LLM-generated synthetic examples and compare performance directly.

Data augmentation and synthetic training data. Wei and Zou [2019] showed that simple rule-based

augmentations improve classification on small datasets. Anaby-Tavor et al. [2020] showed that GPT-2-generated examples improve text classification when labelled data is scarce. Møller et al. [2023] found LLM-generated data most useful as augmentation rather than a full replacement, a finding our results replicate.

3 Data

We use the hotel and restaurant splits of M-ABSA Wu et al. [2025]. Each example is a review sentence paired with (aspect term, aspect category, sentiment polarity) triplets under a fixed domain schema: hotel reviews use categories such as *rooms cleanliness*, *service general*, and *location general*; restaurant reviews use *food quality*, *service general*, and *ambience general*.

The English training set contains 1,255 examples per domain. The Spanish translated set contains the same 1,255 examples machine-translated into Spanish. The Spanish test set contains 584 examples and is used for all Spanish evaluations. The synthetic set (Model C) contains 1,080 examples after filtering; the combined set (Model D) merges translated and synthetic data for 2,335 examples total.

4 Setup

4.1 Experimental Conditions

Table 1 summarises the four training conditions. Three Spanish conditions are each paired with the same English training set; the restaurant split repeats this setup identically. All models use 5 epochs, learning rate 3×10^{-4}, batch size 2, and bf16 precision. A zero-shot Mistral-7B-Instruct-v0.2 (4-bit quantised) baseline is also evaluated.

#	Lang.	Training Data
A	English	Real (1,255)
B	Spanish	Translated (1,255)
C	Spanish	Synthetic LLM (1,080)
D	Spanish	Combined (2,335)

Table 1: Experimental conditions. Spanish test sets use the M-ABSA translated split ($n = 584$).

4.2 Synthetic Data Generation

Synthetic Spanish data was generated using Claude Sonnet 4.6. Rather than prompting from a category list, the full real training split was passed as few-shot context so the model could infer the label set, sentence style, and triplet structure directly from the data. Aspect terms were required to appear verbatim in the sentence, and categories were constrained to those already present in the training set.

Prompt to Claude Sonnet 4.6

```
I am going to paste a Spanish ABSA
training file in M-ABSA format.  Each
line is:
<sentence>####[['aspect_term',
'aspect_category', 'sentiment'], ...]

Sentiment is always one of:  positive,
negative, neutral.  Aspect terms must
be words that appear verbatim in the
sentence.  Categories must be exactly
as they appear in the examples.  Do
not invent new ones.

[training data appended here]

Generate 200 new Spanish examples
in exactly the same format.  Match
the style, vocabulary, and category
distribution of the examples above.
Each sentence should have 1 to 3
triplets.  Vary sentence length, tone,
and domain.  Do not repeat sentences
from the training data.  Output only
raw lines, no numbering, explanations,
or markdown.
```

Each batch of 200 examples was validated to discard lines that did not parse, contained unseen categories, or had no valid triplets. Four batches were generated per domain, yielding ≈270 clean examples per batch.

5 Results

We report micro-F_1 for exact triplet matches: a prediction is correct only if all three elements match the gold label exactly. A prediction with the right aspect term and sentiment but a wrong category scores zero.

5.1 Hotel Results

Table 2 shows results for all hotel conditions. Model D (combined) is the strongest Spanish fine-tuned system ($F_1 = 46.84$), outperforming the translated-only baseline by 3.64 points. Model C (synthetic only) falls far below all other fine-tuned conditions. Zero-shot Mistral-7B is weakest in both languages.

Condition	P	R	F_1
A – EN real	60.71	46.21	52.48
A – cross-lingual	44.74	3.85	7.09
B – ES translated	50.76	37.60	43.20
C – ES synthetic	17.83	15.06	16.33
D – ES combined	51.07	43.26	**46.84**
B – reverse xling	32.26	22.65	26.61
D – reverse xling	31.52	25.59	28.25
Mistral-7B ES	7.05	1.81	2.88
Mistral-7B EN	7.56	5.78	6.55

Table 2: Precision, recall, and micro-F_1 for hotel conditions.

5.2 Restaurant Results

Table 3 shows the same ordering on the restaurant split. Model D achieves the best Spanish result ($F_1 = 44.23$). The pattern across all conditions mirrors the hotel split, confirming that the main findings hold across domains.

Condition	P	R	F_1
A – EN real	55.17	43.69	48.77
A – cross-lingual	32.86	24.67	28.18
B – ES translated	48.89	37.60	42.51
C – ES synthetic	31.08	23.09	26.50
D – ES combined	49.51	39.97	**44.23**

Table 3: Precision, recall, and micro-F_1 for restaurant conditions.

6 Analysis

6.1 Per-Model Performance

Model A (English ceiling). F_1 of 52.48%, with precision exceeding recall (60.71% vs. 46.21%). The model is conservative and tends to under-generate on sentences with multiple aspects.

Model B (translated baseline). F_1 of 43.20%, a drop of 9.28 points from Model A. Translation noise is the likely cause: aspect term boundaries shift, sentiment-bearing phrases change in rendering, and input-output alignment degrades. Machine translation remains viable but the quality loss is visible in the results.

Model C (synthetic only). F_1 of 16.33%, the weakest fine-tuned result. Low precision (17.83%) and recall (15.06%) indicate a failure to learn the task structure. The cause is distributional mismatch between LLM-generated training text and machine-translated test reviews, as shown by the early-stopping ablation in Section 6.6.

Model D (combined). F_1 of 46.84%, outperforming Model B by 3.64 points. Recall improves from 37.60% to 43.26% with little change in precision, so synthetic examples help the model find more correct triplets without adding false positives. Augmentation works; full replacement does not.

6.2 Cross-Lingual Transfer

Model A scores only $F_1 = 7.09\%$ on Spanish despite precision of 44.74%, with recall at 3.85%, meaning the model almost never produces a prediction on Spanish input. Reverse transfer (Spanish to English) is stronger at 26–28%, consistent with mT5's English-heavy pre-training: Spanish representations sit closer to the English space and transfer more readily in that direction.

6.3 Zero-Shot Mistral-7B

Mistral-7B scores 2.88% on Spanish and 6.55% on English, below every fine-tuned model. The main failure mode is producing no output for multi-aspect reviews, which accounts for most of the recall gap. A secondary issue is aspect boundary errors, where adjectives are extracted as standalone terms rather than the nouns they modify. Fine-tuning mT5-small (300M parameters) on a few hundred in-domain examples scores higher than zero-shot prompting of a 7B model for this task.

6.4 Generalisation to Restaurant Reviews

Figure 2 shows the restaurant results. The ranking across conditions is identical to the hotel split: Model A strongest, Model B the best Spanish-trained condition, Model C weakest, and Model D outperforming Model B. Restaurant reviews are slightly harder because categories like *food quality*, *service general*, and *ambience general* frequently co-occur in the same sentence, increasing category confusion and missed triplets. The B-to-D gain is smaller on restaurant (1.72 points) than hotel (3.64 points), but the direction is consistent across both domains.

6.5 Error Analysis

We sampled 30 incorrect predictions per model and classified each error (Table 4). Missed triplets are the most common error type (43–52% across all models). Model A has the highest rate of category confusion (15/48, 31%), showing that assigning the

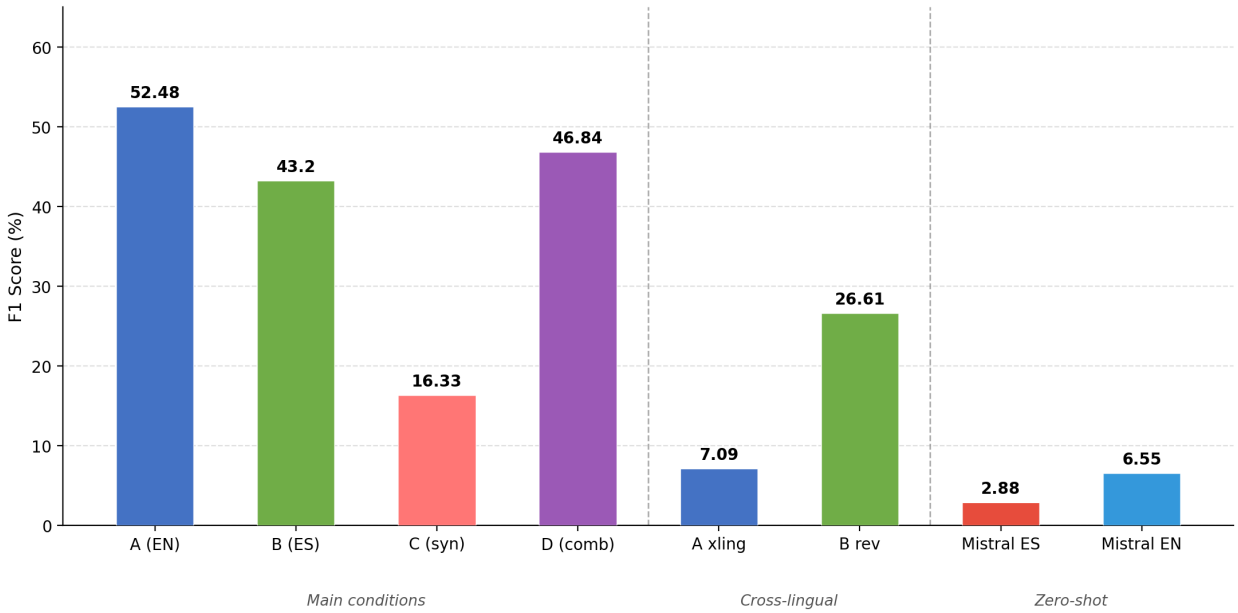

Figure 1: F_1 scores across all hotel experimental conditions. Model D (combined) is the only Spanish fine-tuned condition to exceed the translated baseline (B). Zero-shot Mistral-7B falls below all fine-tuned models in both languages.

correct category is harder than locating the aspect term, regardless of language. Model C has the most evenly distributed error profile, consistent with broad distributional mismatch rather than one dominant failure mode. Model D has the fewest missed triplets, matching its higher recall.

Error Type	A	B	C	D
Missed triplet	25	22	21	18
Category confusion	15	10	12	10
Wrong aspect term	6	10	10	11
Wrong sentiment	1	4	5	3
Spurious prediction	1	0	0	0
Total	48	46	48	42

Table 4: Error type counts (sample of 30 incorrect predictions per model).

6.6 Training Dynamics and Early-Stopping Ablation

Figure 3 shows training and validation loss per epoch. Model C reaches unusually low validation loss by epoch 3 (0.234), well below the other models. To test whether this reflected damaging overfitting, we trained a 3-epoch variant of Model C and evaluated it on the hotel split: it scored $F_1 = 12.87$ ($P = 13.87$, $R = 12.00$), lower than the 5-epoch result of 16.33. Early stopping makes performance worse, not better. More training on synthetic data helps the model learn the output format; the ceiling is the distributional gap between LLM-generated training text and machine-translated test reviews, not over-training.

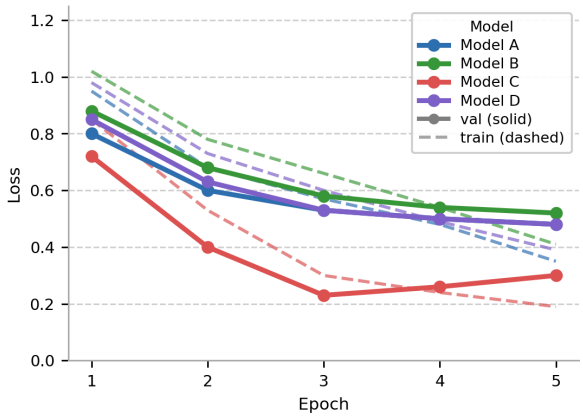

Figure 3: Training (dashed) and validation (solid) loss per epoch for Models A·D. Model C·s val. loss flattens early (see §6.6).

6.7 Qualitative Comparison

Table 5 compares model predictions on a matched sentence pair. The gold labels are `(bathroom, rooms_cleanliness, neg)` and `(staff, service_general, pos)`. Model D is the only system to get both triplets right from English. On Spanish it recovers the service triplet but misclassifies the bathroom category. This is likely a frequency effect: *service general* appears far more often in training than *rooms cleanliness*, so the model defaults to it when surface cues are weak.

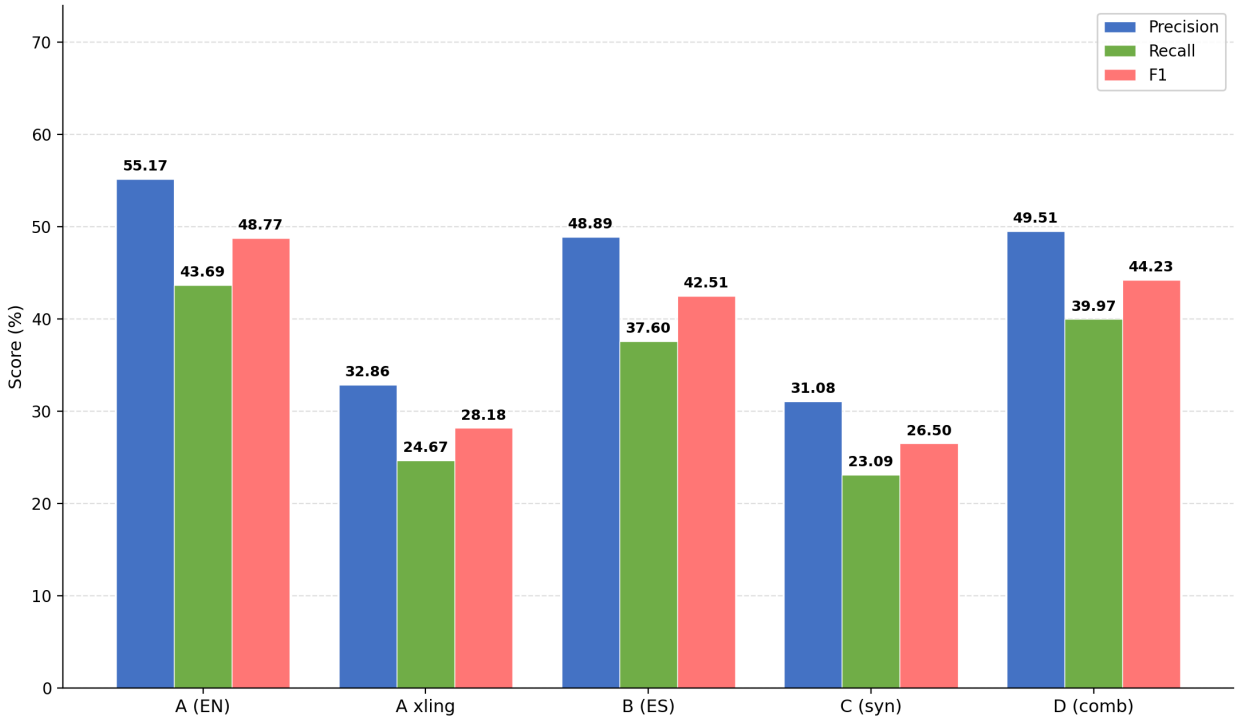

Figure 2: Restaurant split results. Model D is the strongest Spanish setup, whilst Model C remains below the translated baseline. The ordering mirrors the hotel split, confirming the main finding holds across domains.

Model	Lang.	Target Triplets (Ground Truth)	Model Predictions	Observations
A	**EN**	1. (bathroom, rooms, neg) 2. (staff, service, pos)	(staff, service, pos)	**Low Recall: Missed the** bathroom triplet entirely.
B	**ES**	1. (baño, rooms, neg) 2. (personal, service, pos)	(habitaciones, rooms, neg)	**Term Swap: Used** "habitaciones" not "baño".
C	**ES**	1. (baño, rooms, neg) 2. (personal, service, pos)	(baño, service, neg)	**Category Confusion:** Mapped bathroom to "service" category.
D	**ES**	1. (baño, rooms, neg) 2. (personal, service, pos)	1. (baño, rooms, neg) 2. (personal, service, pos)	**Correct: Both triplets** extracted successfully.

Table 5: Predictions for "The bathroom was disgusting but staff were helpful" and its Spanish translation.

7 Conclusion

Synthetic data generated by Claude Sonnet 4.6 does not match machine-translated training data on its own ($F_1 = 16.33$ vs. 43.20 on the hotel split), but adding it to the translated set improves recall without hurting precision, giving the best Spanish result in both domains ($F_1 = 46.84$ hotels, 44.23 restaurants). The gain comes from recall: Model D finds more correct triplets without adding false positives, showing that synthetic examples help the model handle more varied phrasing at test time.

A 3-epoch early-stopping ablation of Model C on the hotel split ($P = 13.87$, $R = 12.00$, $F_1 = 12.87$) confirms that weak performance is not due to overfitting. Reducing training makes things worse: the model needs more exposure to the synthetic distribution to learn the output format. The real problem is that LLM-generated training text and machine-translated test reviews come from different stylistic distributions, and more training cannot bridge that gap.

Fine-tuning mT5-small (300M parameters) on a few hundred in-domain examples scores higher than zero-shot Mistral-7B across all conditions and both languages, which matters for deployment settings where inference cost and latency are constraints.

Limitations. All experiments use mT5-small with batch size 2, whilst Wu et al. [2025] use mT5-base with batch size 16, so scores are not directly comparable. The 3-epoch ablation was run on the hotel split only. Only two domains and one target language (Spanish) are evaluated, and synthetic data was generated with a single model (Claude Sonnet 4.6), so results may not generalise to other generators or language pairs.

Future work. Three directions follow from these findings. First, aligning the style of synthetic generation with the test distribution, for example by conditioning on machine-translated examples rather than original English, could directly reduce the mismatch identified here. Second, varying the translated-to-synthetic ratio in Model D may yield further gains beyond the 50/50 split used here. Third, applying the approach to lower-resource languages such as Basque or Irish, where high-quality machine-translated training data is itself scarce, would test how far LLM-based synthesis can compensate.

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